

A  
**Project Review Meeting**  
on  
**LLM-Based Customer Support Chatbot with  
Knowledge Retrieval**

Submitted by

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Group: 15 Section : 7

**Specialization-**

APPLIED MACHINE LEARNING FOR TEXT ANALYSIS

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# Outline of the presentation

- Problem Statement
- Objectives
- Literature Survey
- Research Gap Identification
- Innovation
- Novelty
- Creativity
- Feasibility
- Project Plan

# Problem Statement

- Traditional chatbots depend on rules and keywords, limiting their ability to handle complex queries.
- Large organizational knowledge bases make information retrieval difficult and time-consuming.
- Standalone LLMs may generate inaccurate or unsupported responses without organization-specific knowledge.
- Existing systems often lack context awareness and source transparency.
- Therefore, there is a need for a RAG-based intelligent chatbot that provides accurate, contextual, explainable, and reliable customer support.

# Objectives

- **Develop** an intelligent customer-facing chatbot using an LLM for automated customer support.
- **Implement** a RAG system using **embeddings** and **FAISS** to retrieve relevant knowledge from support documents.
- **Generate** accurate and context-aware responses using the retrieved information.
- **Improve reliability and transparency** through **source attribution** and **confidence estimation**.
- **Support multi-turn conversations** using conversation history and contextual understanding.



# Literature Survey

| S. No. | Research Paper                                                                                                          | Year | Method / Technology                           | Key Contribution                                                                                                                                                      | Relevance to Our Project                                                                                                           | Link                             |
|--------|-------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 1      | <b>Retrieval Enhancements for RAG: Insights from a Deployed Customer Support Chatbot</b> — Daniel González Juclà et al. | 2026 | RAG, embeddings, chunking, RRF, LLM reranking | Studies retrieval improvements in a <b>real deployed customer-support chatbot</b> and shows that better retrieval improves downstream response quality.               | Directly supports our <b>embedding, chunking, FAISS retrieval and Top-K document selection.</b>                                    | <a href="#">ACL Anthology</a>    |
| 2      | <b>Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering</b> — Zhentao Xu et al. | 2024 | RAG, LLM, Knowledge Graph, semantic retrieval | Uses structured knowledge from historical customer-service issues to improve retrieval and question answering; reports improved retrieval and generation performance. | Supports our <b>customer-service QA and knowledge retrieval</b> approach and provides a possible future Knowledge Graph extension. | <a href="#">Research Paper —</a> |



# Literature Survey

| S. No. | Research Paper                                                                                                               | Year | Method / Technology                                   | Key Contribution                                                                                                  | Relevance to Our Project                                                                         | Link                          |
|--------|------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------|
| 3      | <b>Supporting Human Operators During Customer Service Interactions with Agentic-RAG</b> — Juan Barrionuevo-Valenzuela et al. | 2026 | Agentic-RAG, retrieval validation, summarization, LLM | Uses RAG to assist human customer-service operators with relevant information, summaries and suggested responses. | Directly supports our proposed <b>human-agent escalation and conversation-context features</b> . | <a href="#">ACL Anthology</a> |



# Research Gap Identification

| Paper                                                                                                                                    | What the Paper Addresses                                                                                                                                                                                     | Identified Gap / Limitation                                                                                                                                                                                                                                                            | How Our Project Addresses the Gap                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Juclà et al. (2026)</b> — <i>Retrieval Enhancements for RAG: Insights from a Deployed Customer Support Chatbot</i>                 | Focuses mainly on improving <b>RAG retrieval quality</b> through embedding comparison, chunking, RRF, and LLM-based reranking in a deployed customer-support chatbot.                                        | The study primarily focuses on <b>retrieval optimization</b> . There is scope to integrate retrieval with a broader customer-support system involving conversational context, explainability, confidence estimation, and intelligent escalation.                                       | Our project combines <b>semantic retrieval with LLM response generation, multi-turn conversation, source attribution, confidence estimation</b> .                                                                                           |
| <b>2. Xu et al. (2024)</b> — <i>Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering</i>         | Uses <b>RAG + Knowledge Graphs</b> to preserve relationships within and between historical customer-service issues and improve retrieval and answer generation.                                              | The approach relies on <b>knowledge-graph construction from historical issue tickets</b> , which adds structural complexity. There is scope for a more lightweight document-based RAG architecture suitable for broader customer-support knowledge such as FAQs, manuals and policies. | Our system uses <b>FAISS</b> to retrieve relevant information from diverse support documents, followed by LLM-based response generation.                                                                                                    |
| <b>3. Barrionuevo-Valenzuela et al. (2026)</b> — <i>Supporting Human Operators During Customer Service Interactions with Agentic-RAG</i> | Develops an <b>Agentic-RAG copilot</b> that assists human operators through summaries, relevant information and suggested responses, using agents for routing, retrieval validation and response generation. | The primary focus is <b>supporting human operators in call centers</b> , rather than providing a complete self-service chatbot with user-facing explainability, confidence information and broader knowledge-base interaction.                                                         | Our project focuses on a <b>direct customer-facing chatbot</b> , while retaining human escalation for unresolved or low-confidence queries. It also proposes <b>source attribution, confidence estimation and multi-turn conversation</b> . |

# Innovation

- RAG-Based Knowledge Retrieval:** Retrieves relevant information from FAQs, manuals, policies, and support documents before generating responses.
- LLM-Powered Responses:** Uses an LLM such as Gemini or Llama 3 to generate natural, accurate, and context-aware answers.
- Semantic Search:** Uses embeddings and **FAISS vector search** to understand the meaning of customer queries rather than relying only on keywords.
- Explainable Responses:** Provides **source attribution and confidence estimation**, helping users understand and trust the generated answers.



# Novelty

- **Unified AI Customer Support:** Integrates **RAG**, **LLM**, **semantic search**, and **conversational AI** into a single platform.
- **Knowledge-Grounded Answers:** Generates responses based on **retrieved organizational knowledge**, reducing unsupported answers.
- **Explainable Responses:** **confidence information** with the generated answer.

# Creativity

- Natural Conversational Interaction:** Allows customers to ask questions in their own words instead of following predefined options.
- Context-Aware Conversations:** Uses previous messages to understand follow-up questions and maintain conversation flow.
- Conversation Summarization:** Summarizes previous interactions to help maintain context and support efficient human-agent handover.

# Feasibility

- Technical Feasibility:** The system can be developed using existing technologies such as React, FastAPI, embeddings, FAISS, and LLMs such as Gemini or Llama 3.
- Data Feasibility:** The chatbot can use available FAQs, product manuals, user guides, policies, and customer-support knowledge bases as its knowledge source.
- Implementation Feasibility:** RAG allows the system to use existing LLMs without requiring the development or training of a large language model from scratch.
- Scalability:** The knowledge base can be expanded by adding new documents and updating their embeddings.
- Evaluation Feasibility:** The system can be evaluated using retrieval accuracy, response relevance, answer correctness, semantic similarity, response latency, and user satisfaction.

# Project Plan

## LLM-Based Intelligent Customer Support Chatbot using RAG

